

Hyunseok Seung

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Professional Appointments

2025 – Present **Postdoctoral Research Associate**, *Department of Statistics, University of Wisconsin–Madison*, Madison, WI
Advisor: [Matthias Katzfuss](#)

Education

2025 **Ph.D.** in Statistics, *University of Georgia*, Athens, GA
Advisors: [Jaewoo Lee](#) and [Yuan Ke](#)
Dissertation: *Scalable and Efficient Learning: Algorithmic Advances for Time Series and Deep Neural Models*

2018 **M.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea
Advisor: [Sangun Park](#)
Thesis: *Modified Likelihood Ratio Tests for Extreme Value Distributions*

2016 **B.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea

Publications

Peer-reviewed Conference Proceedings

- C1. **Hyunseok Seung**, Lee, J. & Ko, H. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* in *AAAI Conference on Artificial Intelligence* (2026).
- C2. **Hyunseok Seung**, Lee, J. & Ko, H. *MAC: An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* in *IEEE International Conference on Data Mining (ICDM)* (2025).
- C3. **Hyunseok Seung**, Lee, J. & Ko, H. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* in *IEEE International Conference on Data Mining Workshop (ICDMW)* (2024).
- C4. **Hyunseok Seung**, Lee, J. & Ko, H. *NysAct: A Scalable Preconditioned Gradient Descent using Nystrom Approximation* in *IEEE International Conference on Big Data* (2024).

Journal Articles

- J1. **Hyunseok Seung**, Han, K., Shen, Y. & Ke, Y. *Enhancing COVID-19 Mortality Prediction with Online Autocovariance Change Points Detection*. *Stat* (2026).
- J2. **Hyunseok Seung**, Lee, J. & Ko, H. *Mean Activation Curvature for Scalable Second-Order Optimization in Deep Networks*. *Knowledge and Information Systems*. (Invited journal extension from the IEEE ICDM 2025 best-ranked paper selection) (2026).

- J3. **Hyunseok Seung & Park, S.** Modified Likelihood Ratio Tests for Extreme Value Distributions. *Communications in Statistics–Theory and Methods* **52**, 5742–5751 (2023).

Manuscripts Under Review

- W1. **Hyunseok Seung & Katzfuss, M.** *Scalable Derivative Gaussian Processes via Exact Gradient Reduction* 2026.

Research

- 2025 – Present **Postdoctoral Researcher, Department of Statistics, University of Wisconsin–Madison**
- **Gaussian Processes with Derivatives** (advised by [Matthias Katzfuss](#))
 - Developed a scalable derivative Gaussian process inference method that replaces full-gradient conditioning with lower-dimensional directional derivatives while preserving the exact gradient information required for target-specific prediction.
 - **Hyperspectral Foundation Models** (collaboration with [Townsend Lab](#))
 - Pre-training vision transformer foundation models on hyperspectral data, followed by fine-tuning for downstream trait prediction. Conformal prediction to quantify uncertainty.
- 2022 – 2025 **Research Assistant, School of Computing, University of Georgia**
- **Deep Learning Optimization** (advised by [Jaewoo Lee](#))
 - Developed a curvature-aware zeroth-order optimization method for fine-tuning LLMs, achieving faster convergence and higher test accuracy than state-of-the-art methods, while cutting memory usage by up to 27% compared to MeZO-Adam.
 - Developed scalable second-order optimization methods using activation covariance, improving test accuracy by 3.6% on vision transformers compared to AdamW.
- 2023 – 2024 **Research Assistant, Department of Educational Psychology, University of Georgia**
- **Topic Modeling** (collaboration with [NPDA Lab](#))
 - Analyzed video and text data using automatic speech recognition and topic modeling, collaborating with researchers in mathematics education and psychology.
- 2021 – 2023 **Research Assistant, Department of Statistics, University of Georgia**
- **Time Series Forecasting** (advised by [Yuan Ke](#))
 - Developed hybrid COVID-19 mortality forecasting models. Utilized online autocovariance change point detection to boost model accuracy by 6% and reduce training time by 99% compared to standard rolling-window cross validation.
- 2018 **Associate Researcher, SK hynix Inc.**
- **Wafer Failure Early Detection System** (advised by [Sangun Park](#))
 - Streamlined semiconductor production by identifying key predictors of wafer failure, using statistical models for high-dimensional fabrication data.
- 2017 – 2018 **Research Assistant, Department of Applied Statistics, Yonsei University**

- **Modified Likelihood Ratio Tests** (advised by [Sangun Park](#))
 - Developed modified likelihood ratio test statistics tailored to highly skewed settings to improve sensitivity to tail departures.

Awards & Honors

2019 – 2024	Teaching Assistantship (Stipend), <i>University of Georgia</i>
2018	Industry-Sponsored Scholarship Recipient, <i>SK hynix Inc.</i>
2018	Best Paper Presentation Award, <i>Korean Data and Information Science Society</i>
2017 – 2018	Brain Korea 21 Fellowship, <i>Korean Government-Funded Graduate Program</i>
2015	Honors Award, <i>Yonsei University</i>

Conference Presentations

Oral

- T1. *An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* 2025 IEEE International Conference on Data Mining (Washington, DC, USA). Nov. 2025.
- T2. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* 2024 IEEE International Conference on Big Data (Washington, DC, USA). Dec. 2024.
- T3. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Data and Information Science Society (Pukyong University, Busan, South Korea). May 2018.

Posters

- P1. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* AAAI Conference on Artificial Intelligence (Singapore). Jan. 2026.
- P2. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2025.
- P3. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2024.
- P4. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Statistical Society (Pusan National University, Busan, South Korea). May 2018.

Teaching

University of Georgia

2019 – 2023	Teaching Assistant	
	– STAT6430 Design Analysis Experiments	Spring 2023
	– STAT6315 Statistical Methods for Researchers	Spring 2023
	– STAT8330 Advanced Statistical Applications and Computing	Fall 2022
	– STAT6420 Applied Linear Models	Fall 2022

– STAT4230 Applied Regression Analysis	Spring 2022
– STAT2360 Program and Data Lit using R,	Fall 2021
– STAT4210 Statistical Method,	Spring 2021
– STAT8440 Statistical Inference Bioinformatics	Fall 2020
– STAT6210 Intro to Statistical Methods	Fall 2020
– STAT3110 Intro to Statistics for Life Science	Summer 2020
– STAT3120 Intro to Probability for Life Science	Spring 2020
– STAT6210 Statistical Methods	Fall 2019

Yonsei University

2018	Lecturer	
	– STAT1001 Introduction to Statistics	Fall 2018
2017 – 2018	Teaching Assistant	
	– STAT1001 Introduction to Statistics	Spring 2018
	– STAT1001 Introduction to Statistics	Fall 2017
	– STAT1001 Introduction to Statistics	Spring 2017

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